



Invited Paper: Physics-Driven Real-Time CPS Challenge

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Flashback RTSS 2024: Deadline Missed? Bosch Challenges 2016 & 2017



■ Performance model of a full blown engine management system

Real World Automotive Benchmarks For Free

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The progress and comparability of real-time analysis methods that are applicable to real-world is slowed by the absence of realistic benchmarks, mainly due to intellectual property (IP) concerns. We propose a method that supports the generation of realistic but IP free benchmark sets. Further, we provide the application characteristics of a specific real-world automotive software system.

Keywords—benchmarks, timing analysis, automotive software

I. INTRODUCTION

A large quantity of innovative functionalities in modern automotive systems are realized using significant amounts of software technologies. As a consequence the job of integrating many different applications onto the same target platform has grown to a more and more complex and time consuming task. One important tool for guaranteeing the correctness of the dynamic behavior of the integrated system, especially for the cyber physical parts, is timing analysis.

There exists a large body of work in the domain of worst-case timing (or real-time) analysis. Each method assumes specific application and platform models addressing a subset of existing real-world timing effects. For cases where the application and platform models are simple enough, maximum utilization bounds can be derived to decide whether or not a given system adheres to some predefined real-time constraints. Prominent examples for this kind of analyses are, for instance, the work of Liu and Layland for independent periodic task sets under rate-monotonic fixed priority scheduling on a single core platform [1] or the work of Bertone on outlier deadline first (EDF) scheduling [2]. An overview of extensions on those basic works can be found under [3]. More complex application and platform models are addressed by approaches based on the so-called busy window analysis combined with reasoning about the critical instant as proposed by Lehocsky [4]. As of today there exist proprietary industry strength tools based on that approach, such as SynTAS, that are capable of analyzing most timing effects in current automotive systems. However, the detailed analysis techniques in those tools are not published, and thus, not accessible for the real-time research community.

As a result, there currently exist only few directly applicable tools and approaches that can cope with the complexity of the dynamic behavior in modern automotive systems. Therefore, simulation based methods are very popular albeit time consuming and inherently unsafe, which is unsatisfactory given the potential for front loading with formal analysis techniques.

Due to the introduction of multi-core execution platforms, the risk of divergence between academic research and industrial practice is currently increasing. The reason is the strongly increased problem space for timing analysis induced by multi-core systems.

Extending existing approaches is very challenging since the system structure and the dynamic system behavior of automotive systems is very complex. The reasons are manifold:

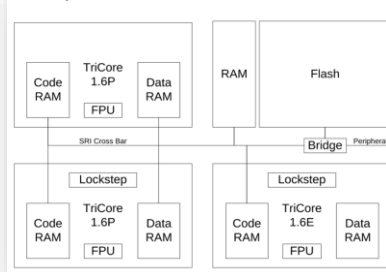
- Control of many physical processes with strongly varying dynamics
- Co-existence of sampled and reactive system parts
- Different time domains (e.g. crank angle, tapers, accounting network traffic) leading to complex scheduling situations
- Numerous complex communication dependencies between functional entities in different time domains due to high coupling which is due to physical dependencies

Sophisticated platform mechanisms influencing the dynamic behavior (cooperative tasks for saving stack space, automated copy mechanisms for data consistency, etc.)

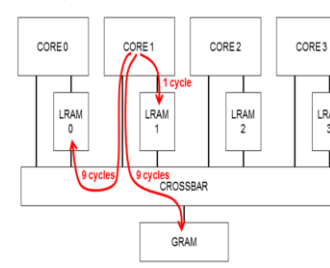
There exist some tools that address the generation of synthetic applications for benchmark purposes that are worth being mentioned here. For instance, the Task Graphs for Free (TGFF) tool [5] generates a set of random independent task trees. Thereby, the structure of the generated task trees is very general, and could more likely be tweaked to fit the structure of automotive applications. This paper can help to do the mapping of the TGFF model to automotive application model including a sensible parametrization. However, TGFF lacks a model for memory accesses that can contribute (depending on the mapping decisions) nontrivially to the execution times especially in multi-core systems.

Another tool for generating synthetic applications models is called System Modeler for Free (SMFP) [6]. The SMFP tool focuses on generating models that are "ready for scheduling analysis". For that purpose, it generates (in contrast to TGFF) not only an application graph, but also a platform graph consisting of computational and communication resources.

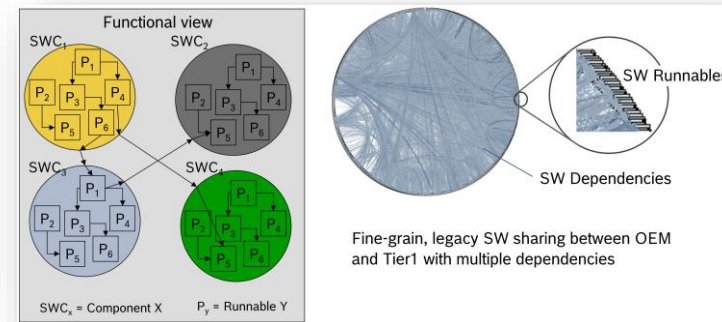
► Simplified AURIX Architecture



► Memory Access Time



HW Model



► Key data of the model

- 1250 Runnables mapped to
- 21 Tasks & Interrupts accessing
- 10.000 Labels (shared data)
- Event chains

	I	II	III	IV	V	VI
	<10	10-50	51-100	101-500	501-1000	>1000

TABLE II. INTER-TASK COMMUNICATION												
Period	1 ms	2 ms	5 ms	10 ms	20 ms	50 ms	100 ms	200 ms	500 ms	1000 ms	1500 ms	2000 ms
1 ms				I	I							
2 ms				I	I							
5 ms				I	I							
10 ms				I	I							
20 ms				I	I							
50 ms				I	I							
100 ms				I	I							
200 ms				I	I							
500 ms				I	I							
1000 ms				I	I							
1500 ms				I	I							
2000 ms				I	I							

► Huge amount of data dependencies

- challenge exact analysis methods

SW Model

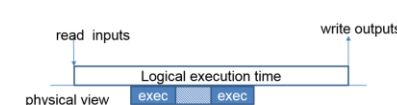
► Implicit Communication

- Goal: data consistency



► LET Communication

- Goal: determinism



Presented at WATERS
Workshop @ ECRTS

Complexity of Cause-Effect Chains

Communication Semantics

Flashback RTSS 2024: Deadline Missed?

Bosch Challenges 2016 & 2017

- Full model amenable to analytical studies capturing real-world problems in 2016 / 2017
- Many fruitful contributions: e.g., end-to-end latency analyses
- The Bosch challenges are still a success if we look at number of citations / downloads

Real world automotive benchmarks for free



Re: The FMTV'16 Challenge
by arne.hamann > Fri Feb 05, 2016

Dear all,

we are finally able to submit the promised Amalthea model [FMTV_Challenge_2016_Bosch_Engine_Control](#) (160.36 KiB) Downloaded 1224 times

The model can be inspected and processed using the Amalthea <http://www.amalthea-project.org/index.php/downloads>.

If you have any questions regarding the model just post it here

Arne

Industrial challenge 2017
by arne.hamann > Thu Feb 16, 2017

We are happy to announce the industrial challenge of WATERS 2017 proposed by Arne Hamann, Simon Kramer, Michael Pressler, Dakshina Dasari, Falk Wurst, and Dirk Ziegenbein from Robert Bosch GmbH.

Details about the intention of the challenge and a high level description can be found here: <https://waters2017.inria.fr/challenge/>.

A detailed description of the actual challenge can be found in the attached document: [WATERS2017_Industrial_Challenge_Bosch.pdf](#) (124.04 KiB) Downloaded 1069 times

The corresponding challenge model file is available in two formats:

The latest version of the Amalthea format which is now supported by a tool chain of the official Eclipse project APP4MC <https://projects.eclipse.org/proposals/app4mc>
[Bosch_ChallengeModel_w_LabelStats_fixedLabelMapping_App4mc_v072.zip](#) (205.5 KiB) Downloaded 677 times

And the previously used Amalthea format supported by a tool chain of the Amalthea4public project <http://www.amalthea-project.org/>
[Bosch_ChallengeModel_w_LabelStats_fixedLabelMapping_Amalthea_v111.zip](#) (205.12 KiB) Downloaded 577 times

- Are there any pressing / significant problems left for those systems?
- **No!** (at least not at Bosch)



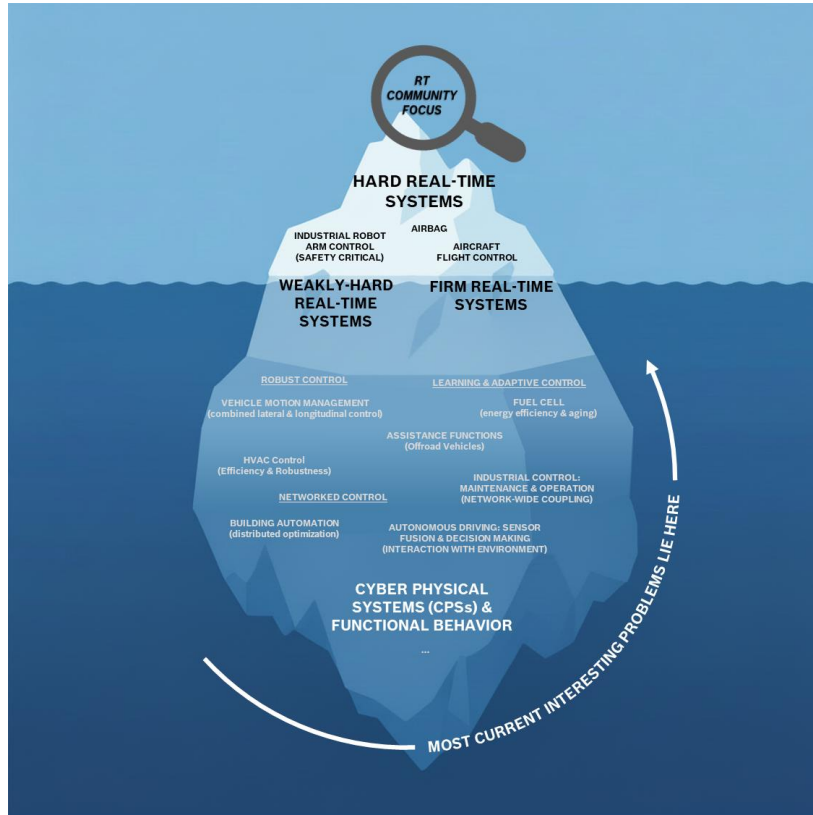
- Performance model of a full blown engine management system

The figure is a collage of various documents and diagrams:

- Top Left:** A paper titled "Real World Automotive Benchmarks For Free" by Simon Kramer, Dirk Ziegenbein, Anne Hamann, Carsten Rensch, Robert Bosch GmbH, Remington, Germany, and simon.kramer@bosch-automotive.com.
- Top Center:** A diagram titled "Simplified AURIX Architecture" showing four cores (CORE0-CORE3) with their respective RAM (Code, Data), FPU, and LRAM components, connected via a Bridge and Lockstep mechanism.
- Top Right:** A diagram titled "Memory Access Time" illustrating memory access cycles between CORE0, CORE1, and shared GRAM.
- Middle Left:** A paper snippet discussing "The progress and comparability of real-time analysis methods..."
- Middle Center:** A large red diagonal stamp reading "OUTDATED".
- Middle Right:** A paper snippet titled "SW Model" discussing communication semantics.
- Bottom Left:** A diagram titled "Function view" showing SWCs (Software Components) and P_r (Runnables) with dependencies.
- Bottom Center:** A diagram titled "Fine-grain, legacy SW sharing between OEM and Tier1 with multiple dependencies" showing SWCs and P_r.
- Bottom Right:** Two diagrams comparing "Implicit Communication" (with copy-in/copy-out actions) and "LET Communication" (with logical execution time).
- Far Right:** A table titled "TABLE II INTER-TASK COMMUNICATION" showing communication metrics across different periods.

Bosch Challenge 2026

The “Hidden Challenges” of the Real-time Ice Berg



- A significant part of real-time systems are **systems controlling some physical plant** – usually closed-loop
- Static real-time models (hard & weakly-hard) as well as established safety & performance metrics **fit basic functional approaches only** (e.g., SISO, settling time, overshoot)
- Modern optimal control-based approaches **consider MIMO systems and inherently incorporate uncertainties** (e.g., Kalman filtering, MPC, RL) to give robustness guarantees
- These systems are currently **industrialized with little contributions from real-time research**

We need new abstractions between the real-time and control engineering

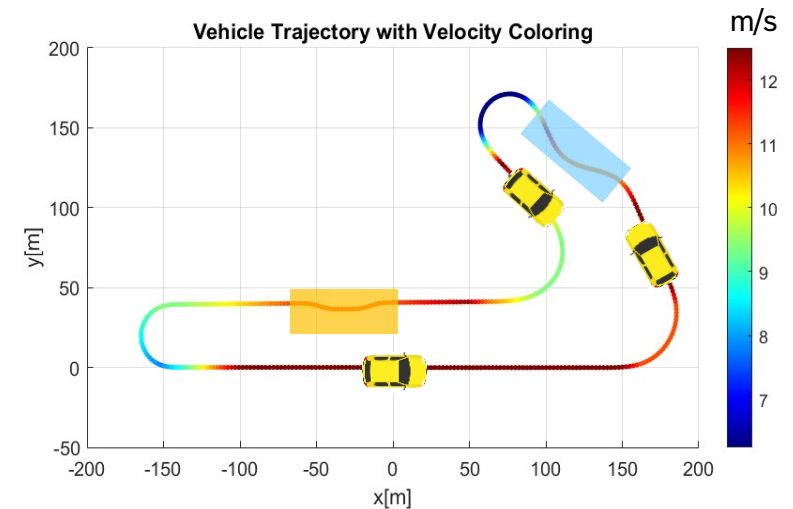
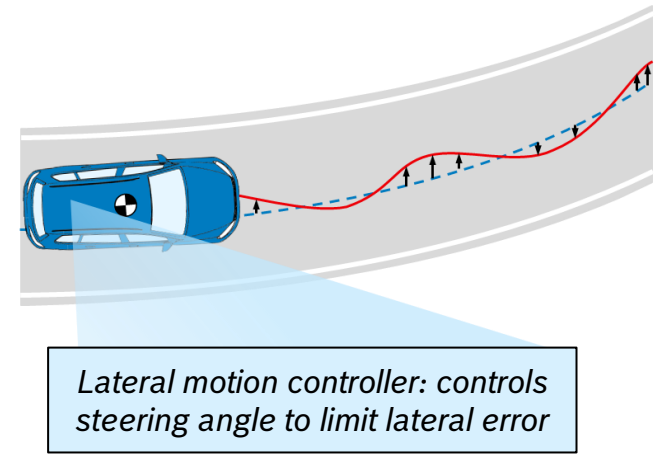
There is a huge opportunity to **build more reliable systems with better performance** if we succeed to integrate real-time considerations into novel control approaches

To fuel that discussion we propose today a
New Bosch Challenge to the community

The New Bosch CPS Challenge

Lateral Motion Control in the Cloud

- Functional application implemented as **distributed task chain**
 - Multiple vehicles on a track, one chain for each vehicle
 - **Controllers share computation and communication resources** in the remote node
 - **Delays and missed updates** might affect the controlled system performance
 - Current interest from Bosch in cloud-based control solutions
- **Not every deadline has the same importance!**
 - Large delays while driving straight or in a curve have different effects → **physical system context matters!**
 - Simple deadline concepts (even weakly-hard constraints) are **insufficient** to fully characterize **context-aware timing sensitivity**
- **Goal:** Linking functional performance (control-related metrics) and temporal performance (real-time deadlines and latencies)



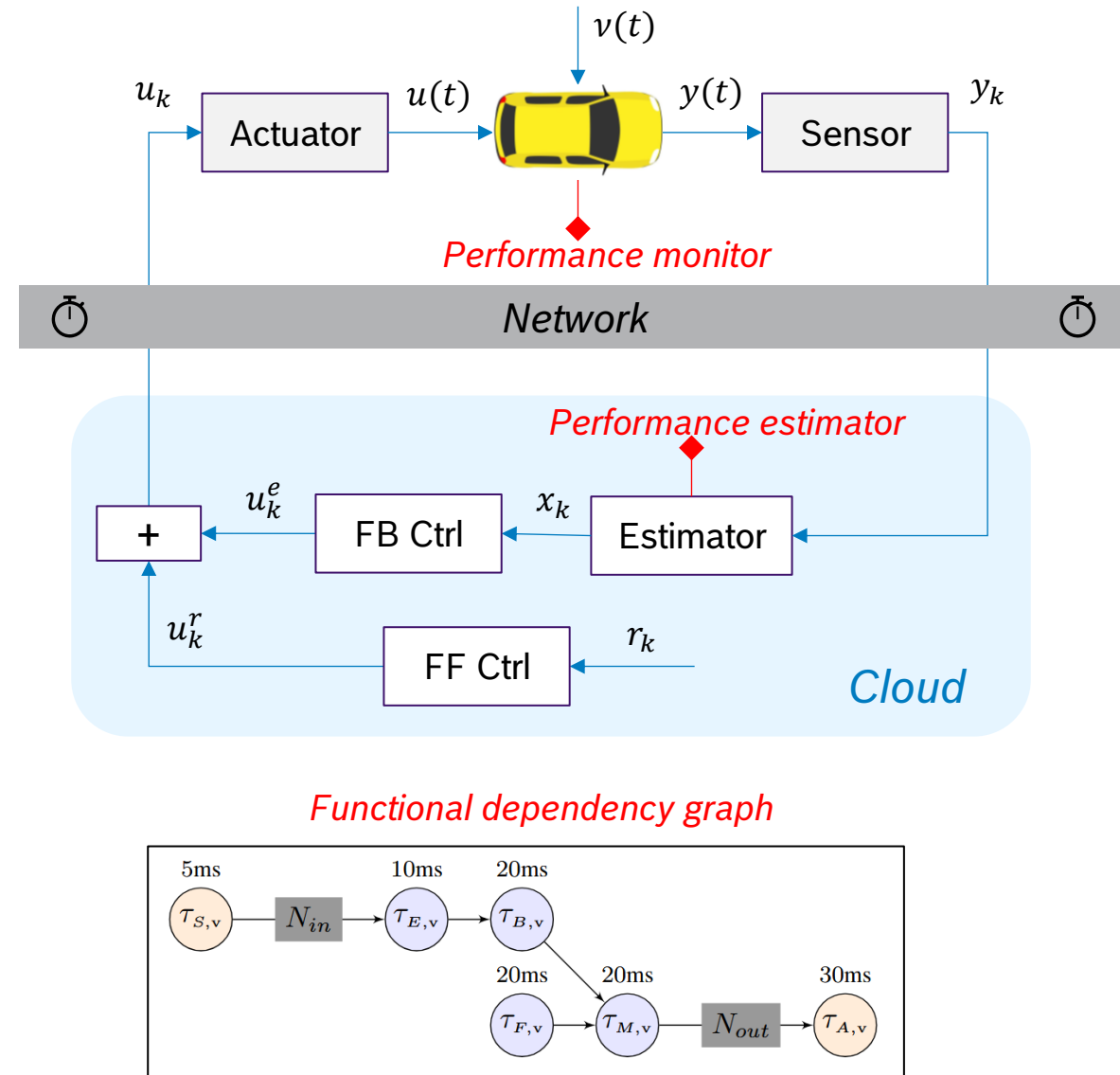
“Critical sections”: double lane change and S-curve at high speed

The CPS Example

Vehicle's Distributed Task Set

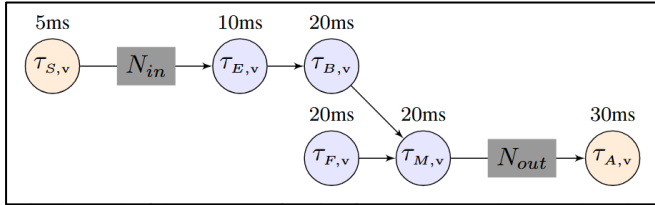
Given design:

- Physical system
- Feedforward + feedback control loop
- Tasks mapped between local and remote (e.g., cloud), with network communication
- All tasks are assumed periodic, with given execution times (distributions)
- Considering read-execute-write or LET protocols
- Performance metrics in-vehicle and in-cloud (estimation)
- Unreliable network: Subject to stochastic delays and possibly packet drops
- Limited number of computational nodes in the cloud, shared between vehicle control tasks



The CPS Example

Design Targets

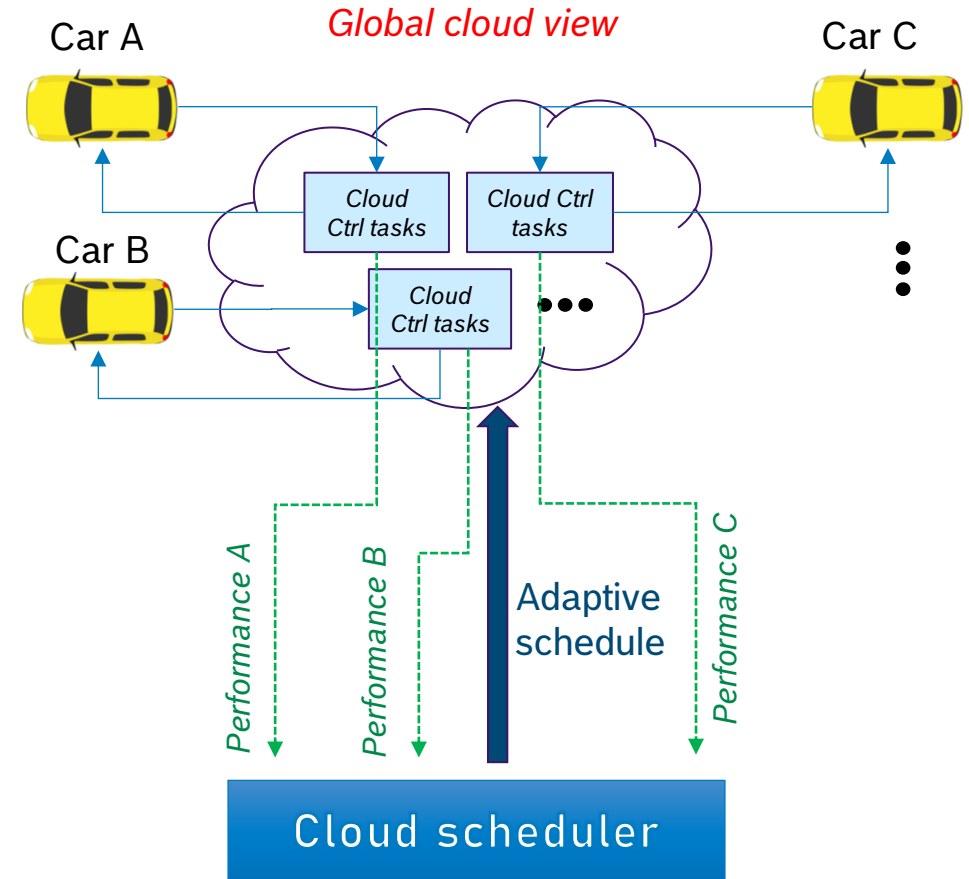


Open design parameters:

- Deadlines, priorities, offsets, communication protocols and **cloud scheduler**
- **The timing layer**, i.e., the scheduling simulator and the triggers of job reads and writes
- Any additional extra load (optional)

Context-Aware Scheduling:

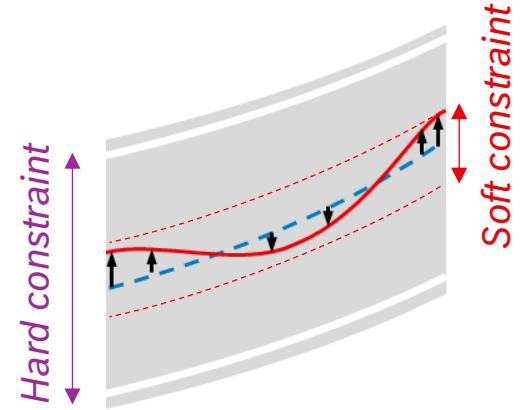
- Using control performance to adjust scheduling mechanism
- E.g., adjust priorities dynamically, so that control tasks for vehicles in critical intervals receive immediate precedence, skipping jobs, graceful degradation



The CPS Example

Performance Metrics and Constraints

- Performance is checked periodically for each vehicle
 - **Safety (Hard Constraint):** Lateral error must **always** be ≤ 0.8 meters
 - **Comfort (Soft Constraint):**
 - Lateral error should be ≤ 0.2 meters
 - **Chance constraint:** Can exceed 0.2 meters for a maximum of **5% of total simulation**
- **Metrics to minimize**
 - **Rolling Performance:** cost of lateral error and its rate of change over a rolling 1-second window
 - **Average Performance:** over the entire simulation
- **Key Considerations:**
 - Solutions ranked based on their success in meeting these constraints and optimization goals
 - Different “difficulty levels” (top speeds) provided



The Simulator FMU structure

Init Parameters (user/test dependent)

- Controller tuning
- Initial state



Provided in the challenge

Inputs

- Trajectory reference
- Current velocity
- Triggers:
job reading/writing
and message
sent/received



Physical model + functional dependencies

Hardcoded parameters

- ❖ Simulation step size
- ❖ Discretized dynamic matrices
- ❖ Performance parameters



Outputs

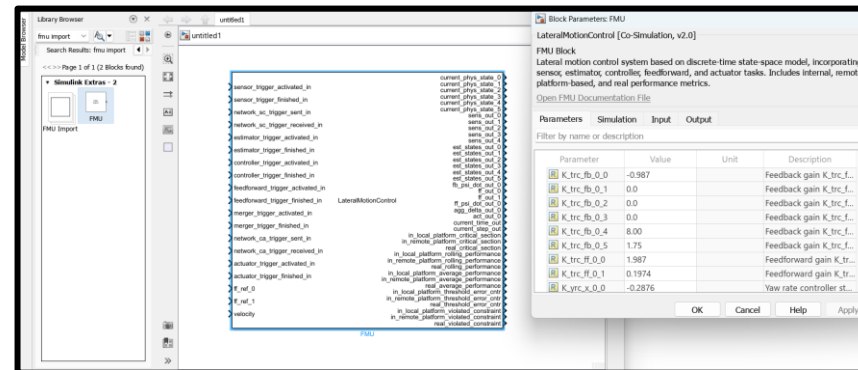
- Current state
- Intermediate variables
- **Performance costs**
- **Critical sections**

To be obtained from scheduler

Can be used as a black box!

Easy to use, e.g., in Simulink

See: <https://fmi-standard.org/>



Targeting the Real-Time Research Community

Challenge Questions

- **Q1: Baseline Behavior:** Maximize the number of vehicles with any *state-of-the-art* scheduling and mapping methods, meeting the performance requirements.
- **Q2: Adaptive Resource/Scheduling:** Maximize the number of vehicles **with *context-aware, adaptive*** in-cloud policies, using runtime performance measurements.
- **Q3: Resource Efficiency:** Minimize cloud computational resources for a *fixed* number of vehicles, while meeting performance requirements.
- **Q4: New Timing Abstractions:** Design *novel timing abstractions* that capture the dynamic relationship between control performance and adaptive timing constraints.
- **Q5: End-to-End Latency:** Analyze resulting *end-to-end latency metrics* (e.g., data age) under various scheduling/communication strategies for control chains.
- **Q6: Data-Driven Strategies:** Propose *context-aware, and adaptive* strategies to solve Q1-Q5, by relaxing periodic task assumptions and assuming *data-driven policies*.

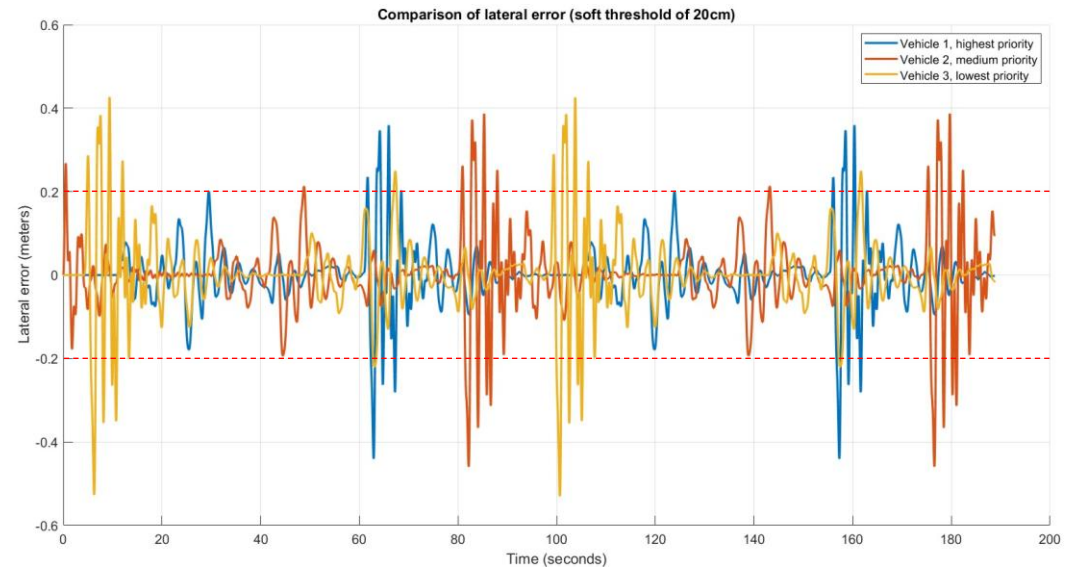
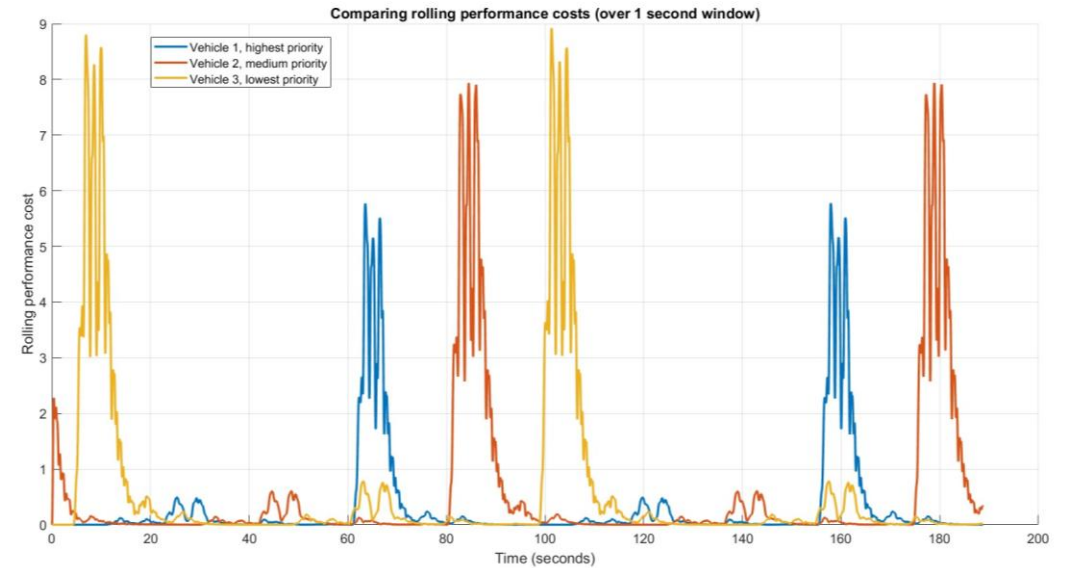
The CPS Example

Example of usage

- **Same end-to-end latency has different effect in different parts of the circuit**
- Adding multiple vehicles to single cloud node
- Standard design with fixed priority to vehicle tasks → each additional vehicle tasks have lower priority
- Each additional vehicle has worse performance
- Vehicle 3 violates 5% threshold!

	Average performance	% of missed soft constraint
Vehicle 1	0.31	3.3%
Vehicle 2	0.50	4.7%
Vehicle 3	0.53	5.7%

- **Need for context-aware priority adaptation** to improve the overall performance



Conclusions

The Challenge in a Nutshell

- **What we provide:**

- A simulator that abstracts a complex controlled cyber-physical system, partially deployed in cloud
- The functional task chain, with example parameters (execution times, periods..)
- Performance monitors of the controlled system and performance metrics to be satisfied

- **What you design:**

- Schedulers and online adaptations driven by the controlled system state and performance monitors
- Obtain the timing layer and the triggers of job reads and writes for the simulation

- **Goals:**

- Create new scheduling policies with control-related insights, aware of the current context (e.g., the current controlled system state)
- Find new abstractions that connect control performance and real-time metrics

- Find the limit of current methods and push that limit forward!

Conclusions

Thank you!

You can find the Challenge at
<https://github.com/boschresearch/CPSChallenge>



- Try it, be inspired, push the limits, give feedback
- This Challenge is not an end product, but a starting point!



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Invited Paper: Physics-Driven Real-Time CPS Challenge

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Abstract—Traditional real-time approaches often ignore the dynamic, state-dependent nature of (functional) requirements in distributed Cyber-Physical Systems (CPSs), leading to overly pessimistic design solutions, which either significantly over-provision computational and communication resources, or result in inferior functional performance. This Challenge aims to foster fundamental discussions and collaborations at the intersection of real-time systems and control engineering. We propose to explore a paradigm shift: moving beyond static, worst-case real-time attributes towards dynamic, context-aware ones, rooted in physics-driven performance. We provide a realistic automotive use-case, featuring a distributed lateral motion control functionality, and challenge the real-time community to develop innovative scheduling and design solutions. The ultimate goal is to facilitate more efficient, robust, and adaptive distributed applications, by aligning real-time system design with the dynamic needs of control, without over-dimensioning resources.

Index Terms—Distributed Control, Real-Time, Cyber-Physical Systems, Industrial Challenge

It should be noted that the impact of each source of uncertainty on the control performance is highly dependent on the system and its operational context. Consequently, timing variations may be considered negligible in certain scenarios, for example, in the case when the sensitivity to physical uncertainties exceeds the one to temporal variations by orders of magnitude, or when the plant exhibits low-pass characteristics, rendering it insensitive to high-frequency timing jitter. Conversely, in other cases, strict and precise timing is of greater importance, for instance, when rapid responses to events are required or if the system exhibits inherent instability.

The above discussion highlights a critical weakness in many current methodologies: While analytically attractive, they often fail to address the complex interplay between the various facets of uncertainty and system dynamics. Consequently, building truly effective and efficient systems requires a paradigm shift.